

Katherine Liberona Irarrázabal

Full stack developer

Montréal, QC

katherine.lib.ira@gmail.com

linkedin.com/in/katherine-liberona-irarrazabal-76a843150

github.com/katherinemli · katherinemli.github.io/Portafolio

About

Full stack developer with 7 years of experience in software, web applications and distributed systems. I work on solutions that combine backend and frontend development, embedded systems, network protocols and data processing. I have helped build and ship solutions across several fields, including satellite communications, network management, logistics and emergency response platforms.

Experience

Comtech Telecommunications · Full stack developer · Montréal, QC

Jun 2022 – present

- **Bare-metal C firmware** for satellite modems (ARM, 3 MB RAM, no OS): SSH (curve25519, ed25519, hardware entropy) replacing Telnet.
- **Satellite network NMS**: Vue.js interface (ag-Grid, Leaflet, ApexCharts) and C services; SNMP integration of network equipment; offline vector maps for low-bandwidth links.
- **Security**: mTLS infrastructure (certificate authority, server and clients), HTTPS and Nginx reverse proxy.
- **Antenna control unit (ACU)** in C on Raspberry Pi 5: points electronically steered antennas (GPS, IMU, modem lock) over the AMIP protocol; sensor calibration.
- **Delivery**: Debian packages, Docker, Jenkins and automated end-to-end tests (Playwright) that replay real customer actions.

wherEX · Full stack developer · Chile

Sep 2020 – Feb 2022

- Led the full migration of a PHP/Angular platform to a microservices architecture with Go and React.
- Designed and built backend services and web interfaces to process and visualize data at scale.
- Implemented a ranking algorithm using user profiles and market data to improve recommendations.
- Built scalable solutions together with engineering and product teams.

Falabella Retail · Full stack developer · Chile

Feb 2018 – Jun 2018

- Designed and built, end to end, a last-mile logistics system for a major retailer, with Go and React.
- Automated key logistics processes, cutting operating costs and optimizing delivery team size by more than 80%.
- Designed and deployed an automated geocoding system processing thousands of addresses a day.
- Optimized delivery routes together with several teams to better meet operational needs.
- Built a predictive system that accurately estimates delivery time windows.

CITIAPS · Front-end developer · Santiago, Chile

Mar 2017 – Jan 2018

- Built a government-backed emergency response platform in Vue.js, with real-time volunteer tracking and mobile notifications.
- Built a Java and AI-based emotion analysis system processing data from multiple cameras.
- Created interactive data visualizations with Vue.js and D3.js for geolocated data analysis.
- Built an interactive route-tracking system for a papal visit with Vue.js and the Google Maps API.

Universidad de Santiago de Chile · Lecturer

Mar 2020 – Mar 2022

- Theoretical and practical foundations of logical thinking and algorithms for engineering students: programming fundamentals, distributed systems.

Technical skills

Backend	Go, Python, C, PHP, Django
Frontend	React, Vue.js, Angular, D3.js
Systems	Linux, embedded systems, multithreaded programming, memory management, network protocols
Data	MySQL, MongoDB, real-time processing
DevOps	Docker, AWS, Nginx, Git, Jenkins
Languages	Spanish, English, French

Projects

- **Orbit Tracker** – daemon, API and real-time dashboard for a tracking mount. [github](#)
- **Route Optimizer** – Go API, shortest route with goroutines. [github](#)
- **Hantavirus Tracker** – live map of geolocated cases. [github](#)